

BI Architecture and Lifecycle: Comprehensive Exam Reviewer

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1. Core BI Concepts & Importance

Definition of Business Intelligence (BI)

- **What it is:** The software, data tools, infrastructure, and best practices used to transform raw data into actionable business decisions.
- **The Core Knowledge Continuum:**

Data → Information → Knowledge → Plans (Action)

- **Components Included:** Data Warehousing, Business Analytics, and Knowledge Management.

Primary Purpose & Business Value

BI allows companies to view **present and historical data** in business context to drive profitability and efficiency.

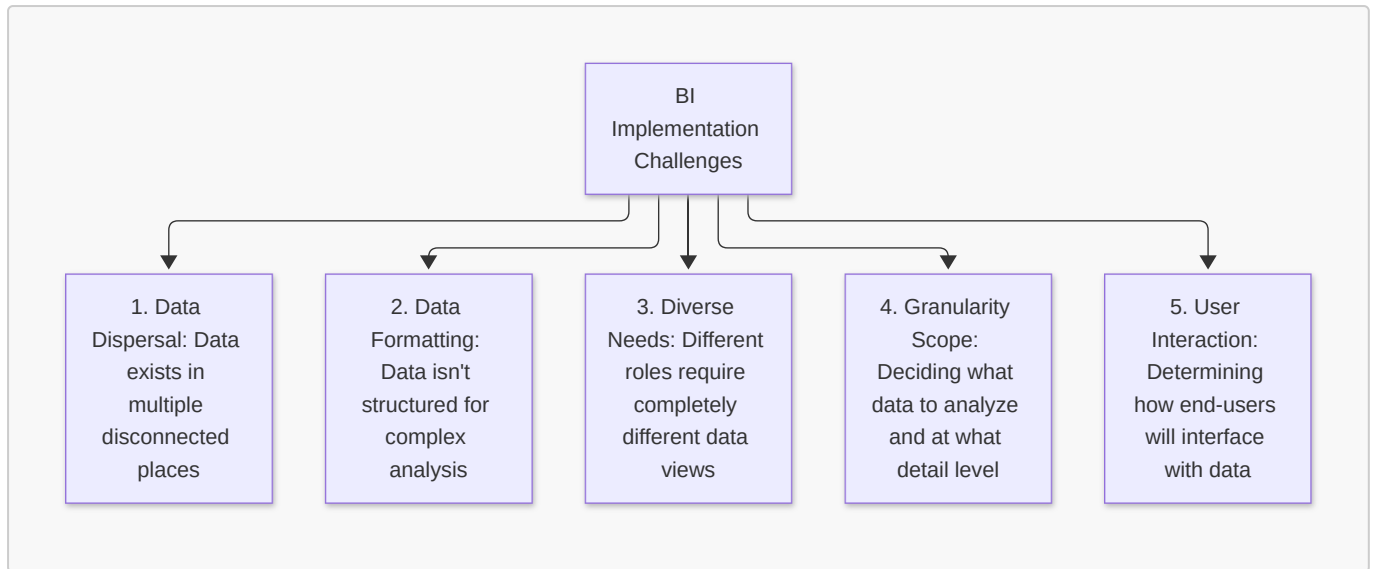
- **Key Benefits:**
 - Identify ways to increase profit
 - Analyze customer behavior
 - Compare internal data with competitors
 - Track performance against metrics
 - Optimize business operations
 - Predict success and spot market trends
 - Discover root causes of operational issues

2. Industry Use Cases

Industry	Primary BI Applications & Benefits
Customer Service	Unified data sources combine customer info and product details so agents resolve questions and issues quickly.
Finance & Banking	Combines customer history and market conditions to evaluate health, detect risk, predict success, and analyze branch performance.
Healthcare	Patient self-service answers, real-time inventory tracking, and operational efficiency without burdening staff.
Retail & Insurance	Benchmarks performance across stores/channels; tracks insurance claims processes to correct missed service targets.
Sales & Marketing	Unifies promo, pricing, sales, and customer action data for targeted customer segmentation and campaign planning.
Security & Compliance	Centralized dashboards isolate root causes of security issues and simplify regulatory reporting.
Supply Chain	Uses a Single Pane of Glass (SPOG) to manage global data, speed up goods movement, and resolve bottlenecks.

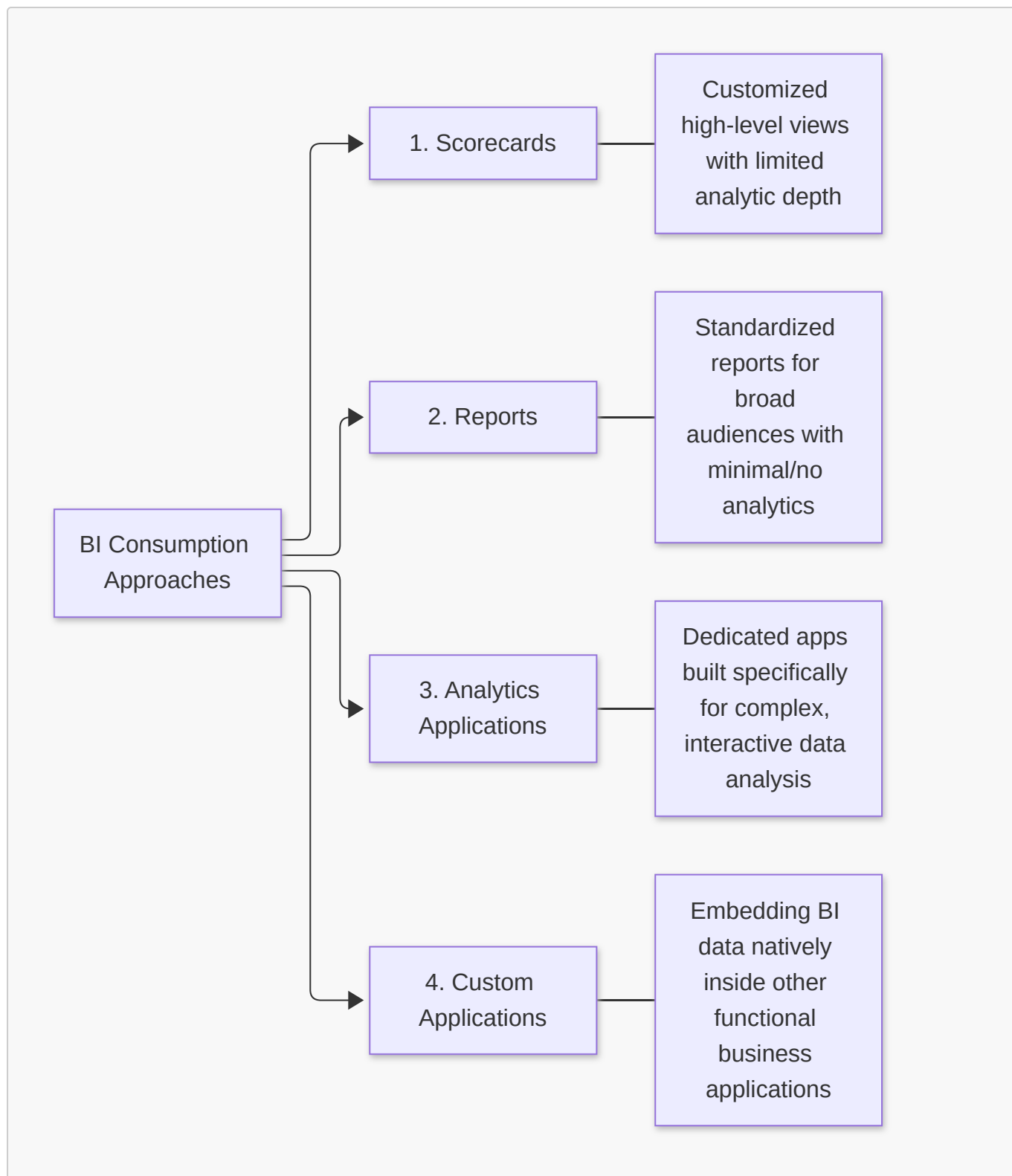
3. Challenges in Building BI Solutions

Creating an effective BI platform requires overcoming five main architectural and design hurdles:



4. Approaches to Consuming BI

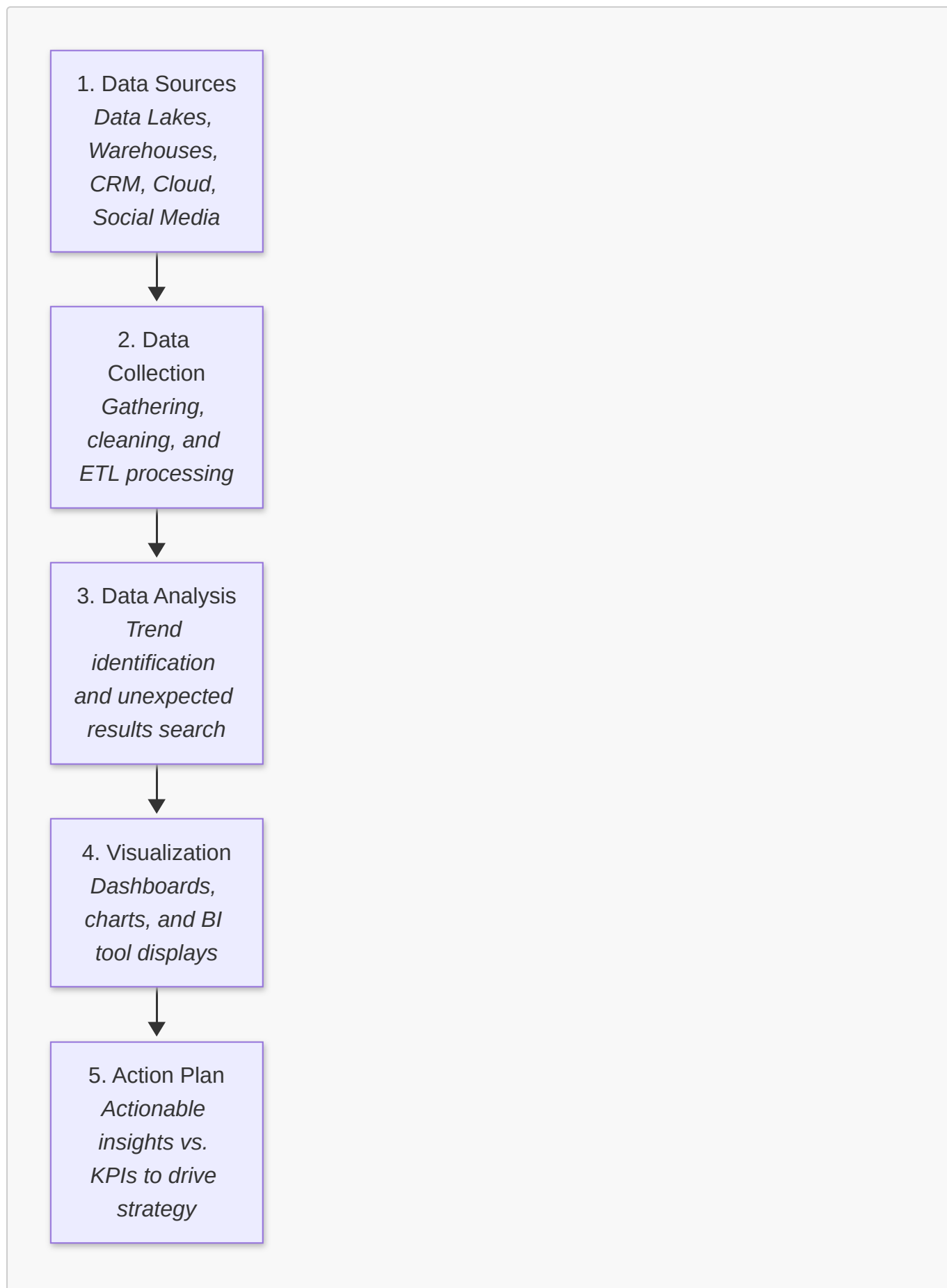
Different users require different interaction styles with BI data.



5. BI Architecture & Data Flow

The 5 Architectural Layers

BI architecture processes raw inputs into actionable decisions across five sequential stages:



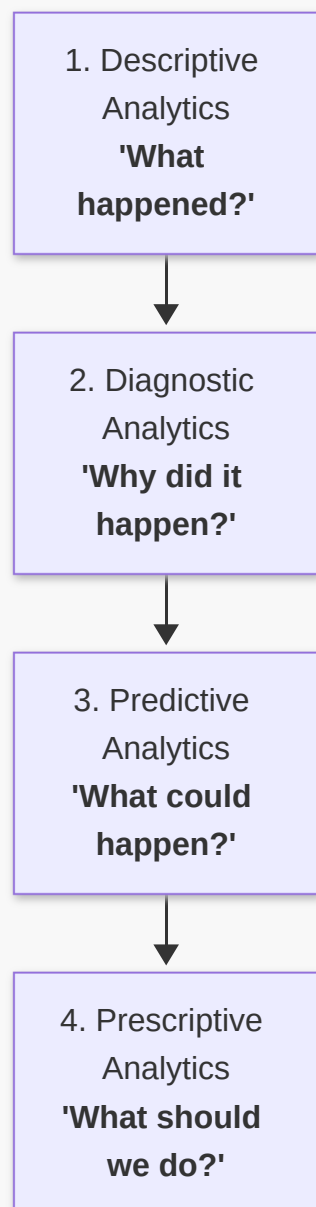
6. The Analytics System Framework

Building a complete analytics environment requires balancing **6 core pillars**:

1. **Business & Quality Context:** Understanding overall organizational performance goals so analytics directly inform decision-making.
 2. **Stakeholders & Users:** Identifying all individuals or groups impacted by, using, or interested in the analytical solution.
 3. **Processes & Data:** Managing raw data materials alongside the business workflows tightly coupled with them.
 4. **Tools & Techniques:** Ensuring software matches the technical needs of data builders and the usability needs of end-users.
 5. **Team & Training (PEOPLE): The most critical component.** Focuses on recruiting, retaining, and upskilling talent.
 6. **Technology & Infrastructure:** Evaluating hardware capacity (servers, networks, storage) to prevent bottlenecks under analytic load.
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7. The 4 Types of Data Analytics

Analytics projects fall into four distinct categories based on complexity and strategic value:



- **1. Descriptive Analytics**
- **Question answered:** What happened in the past?
- **Function:** Examines historical data to baseline trends (e.g., calculating average monthly sales for the past year).
- **2. Diagnostic Analytics**
- **Question answered:** Why did it happen?
- **Function:** Deep-dives into datasets to compare variables and isolate root causes (e.g., investigating why sales dropped in a specific month).
- **3. Predictive Analytics**
- **Question answered:** What could happen in the future?

- **Function:** Applies historical data patterns to forecast future outcomes (e.g., forecasting next quarter's revenue).
- **4. Prescriptive Analytics**
- **Question answered:** What action should be taken?
- **Function:** Recommends specific strategies to exploit predicted outcomes (e.g., recommending optimal marketing spend allocation to maximize upcoming sales).

8. Analytics System vs. Technology System

Exam questions frequently test the distinctions between these two systems:

Aspect	Analytics System	Technology System
Primary Focus	Provides actionable insights for decision-making via data analysis, models, and statistical methods.	Provides infrastructure and tools needed to store, process, and manage data.
Key Components	Context, Stakeholders, Processes, Tools, and Skilled People/Training .	Hardware (servers, networks), Software (databases), ETL pipelines, and Integrated Platforms.
Role of Data	Raw material converted into insights and predictive models.	Materials transformed, transported, and stored across infrastructure.
Primary Output	Insights, dashboards, statistical models, strategy recommendations.	Infrastructure stability, high system availability, security, fast throughput.
Primary Users	Business Analysts, Data Scientists, Executive Decision-makers.	IT Engineers, System Administrators, Database Administrators (DBAs).
Concrete Examples	Predictive models in Python/R; dashboards in Power BI or Tableau.	SQL Server, Oracle DB, Cloud (AWS, Azure), ETL Data Pipelines.

9. Quick-Recall Exam Prep & Cheat Sheet

Common Trap Questions & Key Distinction Rules

- **Single Pane of Glass (SPOG):** A supply chain term for consolidating global operations data onto a single screen to locate bottlenecks.
- **People First Rule:** If asked to pick the **most important consideration** in an analytics framework, the answer is always **People / Team & Training**.
- **Data Flow Progression:** Remember the exact order: Sources → Collection → Analysis → Visualization → Action Plan.
- **System Difference Shortcut:**
- **Technology System** = *Hardware/Software/Storage* (Built by IT to run reliably without continuous human design changes).

- **Analytics System** = *Insights/Models/People* (Built for business decision-makers to answer strategic questions).